CAD/CAM LAB

Course Code: 23A03602P

B.Tech (ME) - III Year II Semester - JNTUA R23 Regulations

Professional Core | L-T-P-C: 0-0-3-1.5

LAB MANUAL

Name	
Roll No.	
Branch / Section	Mechanical Engineering
Academic Year	

Course Objectives

1. Develop skills in drafting and understanding orthographic and isometric views, and CAD file formats like DXF and IGES.
2. Enable the creation of 3D part models using basic and advanced features in CAD tools.
3. Hands-on experience in assembly modeling using both feature-based and Boolean-based methods.
4. Familiarize with CAM software for generating NC code for various machining processes.
5. Expertise in industrial manufacturing via post-processors and NC machines for real-time machining.

Course Outcomes

CO	Outcome	Level
1	Create accurate 2D technical drawings using orthographic and isometric projections, and interpret file formats like DXF and IGES.	L1, L2, L3
2	Develop 3D part models using protrude, revolve, shell and sweep, and demonstrate parent-child relationships.	L2, L3, L5
3	Construct and analyze assemblies using feature-based and Boolean operations; create complex mechanical assemblies.	L3, L4, L6
4	Generate NC code for complex surfaces using CAM tools and study the function of various post-processors.	L1, L3, L6
5	Transfer NC code to CNC machines; perform and evaluate basic turning and milling tasks.	L4, L5, L6

List of Experiments

1. **Drafting:** part drawings (orthographic and isometric), dimensioning and tolerances, scanning and plotting; study of script, DXF and IGES files.
2. **Part Modelling:** 3D models through protrusion, revolve, shell, sweep; creation of various features; study of parent-child relation.
3. **Assembly Modelling:** feature-based and Boolean-based modelling surfaces; assembly modelling and design of simple components.
4. **CAM:** (a) study of post processors used in NC machines; (b) NC code for free-form and sculpted surfaces using CAM packages; (c) machining of simple components on NC lathe and mill by transferring NC code / from a CAM package.
5. **Evaluation of Stress/Strain** for a plate with a hole.

Experiments are to be done through any four software packages from: AutoCAD, MicroStation, CATIA, Pro-E, I-DEAS, CAEFEM, Gibbs CAM, Master CAM, etc.

Experiment 1: Drafting

Aim

To prepare 2D part drawings (orthographic and isometric views) with dimensions and tolerances, to plot them, and to study script, DXF and IGES files.

Software

AutoCAD (or any 2D/3D CAD package such as CATIA / Pro-E drafting module).

Theory

- **Orthographic projection:** front, top and side views projected on mutually perpendicular planes (first-angle method is used in India).
- **Isometric view:** a pictorial view where the three axes are 120 degrees apart and all lengths along the axes are drawn to the same scale. In AutoCAD, set the snap style to Isometric.
- **Dimensioning:** linear, aligned, angular, radius, diameter and baseline/continuous dimensions. Tolerances may be unilateral, bilateral or limit type, e.g. 25 +0.02/-0.01.
- **Script file (.scr):** a text file of AutoCAD commands executed in sequence using the SCRIPT command.
- **DXF (Drawing Exchange Format):** ASCII (or binary) file from Autodesk with sections HEADER, TABLES, BLOCKS, ENTITIES, used to exchange 2D/3D drawing data.
- **IGES (Initial Graphics Exchange Specification):** neutral, ASCII file format with Start, Global, Directory Entry, Parameter Data and Terminate sections; carries wireframe, surface and solid data between CAD systems.

Procedure

1. Set up units, limits and layers (object, hidden, centre, dimension).
2. Draw the front view using LINE, CIRCLE, ARC, OFFSET, TRIM and FILLET.
3. Project the top and side views using construction lines; use hidden (dashed) lines for invisible edges.
4. Switch to isometric snap (F5 to change planes) and draw the isometric view.
5. Add dimensions and tolerances using DIMSTYLE settings; add title block.
6. Save as .dwg, then use SAVEAS / DXFOUT to export .dxf, and EXPORT or a translator to get .iges.
7. Open the DXF/IGES file in a text editor and identify its sections.
8. Plot using PLOT command: select the plotter/PDF printer, paper size, scale and plot area (Window/Extents).

Sample AutoCAD script (.scr)

```
LINE 0,0 100,0 100,50 0,50 C
CIRCLE 50,25 10
ZOOM E
```

Result

The orthographic and isometric views were drawn, dimensioned, exported to DXF/IGES and plotted.

Viva Questions

1. Difference between first-angle and third-angle projection?
2. What is the purpose of the DXF format?

3. Name the sections of an IGES file.
4. What is a script file?

Experiment 2: Part Modelling

Aim

To generate 3D models using Protrusion (Extrude), Revolve, Shell and Sweep features, create other features, and study the parent-child relationship.

Software

CATIA / Pro-E (Creo) / SolidWorks or any parametric solid modeller.

Theory

Feature	Description
Protrusion (Extrude / Pad)	Adds material by extruding a closed 2D sketch along a straight path to a given depth.
Revolve	Sweeps a sketch about an axis through a given angle; used for shafts, pulleys and bushes.
Shell	Hollows a solid, leaving walls of specified thickness; selected faces are removed (open).
Sweep	Moves a profile along a trajectory (path) to create pipes, springs and handles.
Other features	Hole, fillet/round, chamfer, rib, pattern (linear/circular), mirror, draft, loft.

Parent-child relationship: a feature created on or referencing another feature is its *child*; the referenced feature is the *parent*. Editing or deleting a parent affects its children (the model regenerates; a child may fail if its reference vanishes). Good modelling keeps references minimal and uses datum planes to avoid unwanted dependencies.

Procedure

1. Start a new part file; select a sketch plane (e.g. Front).
2. Sketch the base profile, constrain and dimension it fully.
3. Apply Protrusion to create the base feature.
4. Create a second sketch on a face; use Revolve to add a cylindrical boss.
5. Use Shell to hollow the part with a wall thickness (e.g. 2 mm).
6. Create a path sketch and a profile; apply Sweep to add a pipe-like feature.
7. Add holes, fillets and patterns.
8. Open the model tree, change a parent dimension, and observe how child features update.

Result

3D models were created using protrusion, revolve, shell and sweep, and the parent-child relation was verified.

Viva Questions

1. What is a parametric model?
2. What happens to child features when a parent is deleted?
3. Difference between Revolve and Sweep?
4. What is a datum plane?

Experiment 3: Assembly Modelling

Aim

To perform feature-based and Boolean-based modelling, assemble simple components, and design simple components.

Theory

- **Feature-based modelling:** the solid is built by adding/removing features (extrude, cut, hole, fillet) in sequence, stored in a history tree.
- **Boolean-based modelling (CSG):** complex solids are built from primitives (box, cylinder, sphere, cone, torus) using **Union** (A U B), **Subtraction/Difference** (A - B) and **Intersection** (A n B).
- **Assembly modelling:** components are positioned using mating constraints: coincident, concentric, parallel, perpendicular, tangent, distance and angle. Approaches: bottom-up (assemble finished parts) and top-down (design parts in the context of the assembly).
- **Degrees of freedom:** a free body has 6 DOF (3 translations + 3 rotations); each mate removes DOF.

Procedure (example: Bolt, Nut and Plate assembly)

1. Model the plate with a hole, the bolt (revolve + hexagonal head) and the nut (hexagon with threaded hole) as separate parts.
2. Open a new assembly file and insert the plate as the fixed (grounded) component.
3. Insert the bolt; apply a concentric mate between bolt shank and plate hole, and a coincident mate between head face and plate face.
4. Insert the nut; apply concentric and coincident mates with the bolt and plate.
5. Run interference check and exploded view; generate a bill of materials.
6. Boolean exercise: create a box and a cylinder, then apply Union, Subtract and Intersect, and compare results.
7. Design a simple component (e.g. a flanged coupling or bracket) using the above methods.

Result

Feature-based and Boolean-based models were created and a simple assembly was constructed and checked for interference.

Viva Questions

1. List the Boolean operations.
2. What is a mate constraint?
3. Bottom-up vs top-down assembly?
4. How many DOF does a rigid body have in space?

Experiment 4: CAM

4(a) Study of Post Processors

A **post processor** converts the neutral tool path data (CLDATA / APT cutter location file) generated by the CAM software into the machine-specific NC program (G and M codes) understood by a particular controller (Fanuc, Siemens, Heidenhain, Mitsubishi, etc.).

- Input: CL file (tool positions, feeds, speeds) + machine definition (axes, limits, controller format).
- Output: NC part program with block numbers, tool changes, coolant, spindle commands and canned cycles.
- Generic posts are customised for axis configuration (3-axis mill, 2-axis lathe, 5-axis) and controller syntax. Examples: Mastercam MP post, CATIA APT post, Gibbs post.
- Wrong post processor selection can cause crashes, so always run a simulation / dry run.

Common G and M codes (Fanuc style)

Code	Function	Code	Function
G00	Rapid positioning	M03 / M04	Spindle CW / CCW
G01	Linear interpolation	M05	Spindle stop
G02 / G03	Circular interpolation CW / CCW	M06	Tool change
G20 / G21	Inch / metric input	M08 / M09	Coolant on / off
G28	Return to reference point	M30	Program end and rewind
G40/41/42	Cutter compensation cancel/left/right	G90 / G91	Absolute / incremental
G71	Rough turning cycle (lathe)	G81	Drilling cycle (mill)
G94 / G95	Feed per min / per rev	G96 / G97	Constant surface speed / RPM

4(b) NC Code for Free-form and Sculpted Surfaces using CAM Packages

1. Import or create the 3D surface model (IGES/STEP/native) in the CAM package.
2. Define stock (block), work coordinate system (G54) and tool library (ball-nose end mill for sculpted surfaces).
3. Select a 3-axis surface strategy: roughing (pocketing / Z-level) followed by finishing (parallel / raster, scallop, or flowline).
4. Set step-over (scallop height), step-down, spindle speed and feed.
5. Simulate the tool path (verify gouges, collisions, remaining stock).
6. Select the correct post processor and generate the NC code.

Because the surface is approximated by many tiny G01 moves, the NC file is very large and is usually transferred to the machine through DNC (RS-232 / Ethernet) rather than typed.

4(c) Machining of Simple Components on NC Lathe and Mill

Sample Program 1 - CNC Turning (Fanuc), finish profile: dia 30 x 40 long, then dia 20 x 20

```
O1001 (TURNING SAMPLE)
N10 G21 G40 G80 G90
N20 T0101 (OD TURNING TOOL)
N30 G96 S120 M03
N40 G50 S2000
N50 G00 X32.0 Z2.0 M08
N60 G71 U1.5 R0.5
N70 G71 P80 Q130 U0.4 W0.1 F0.2
N80 G00 X0
N90 G01 Z0 F0.1
N100 X20.0
N110 Z-20.0
N120 X30.0
N130 Z-40.0
N140 G70 P80 Q130 (FINISH CYCLE)
N150 G28 U0 W0
N160 M05 M09
N170 M30
```

Sample Program 2 - CNC Milling (Fanuc), 60 x 40 rectangular profile and drilling of 4 holes

```
O2001 (MILLING SAMPLE)
N10 G21 G17 G40 G49 G80 G90
N20 T01 M06 (DIA 10 END MILL)
N30 G54 G00 X-10.0 Y-10.0
N40 S1500 M03
N50 G43 H01 Z50.0 M08
N60 G00 Z2.0
N70 G01 Z-5.0 F100
N80 G41 D01 X0 Y0 (CUTTER COMP LEFT)
N90 G01 X60.0 F150
N100 Y40.0
N110 X0
N120 Y0
N130 G40 X-10.0 Y-10.0
N140 G00 Z50.0
N150 T02 M06 (DIA 6 DRILL)
N160 S1000 M03
N170 G43 H02 Z50.0
N180 G81 X10.0 Y10.0 Z-12.0 R2.0 F80
N190 X50.0
N200 Y30.0
N210 X10.0
N220 G80
N230 G28 G91 Z0
N240 M05 M09
N250 M30
```

Procedure (CAM package route)

1. Create the part geometry and define stock and tools.
2. Generate tool paths (turning / contour / drilling).
3. Simulate, post-process, and save the NC file.
4. Transfer to the CNC controller via DNC or memory card; set work offsets and tool offsets on the machine.
5. Run in dry-run / single-block mode, then machine the component.

Result

NC codes were generated for free-form surfaces and simple components, and turning and milling operations were carried out/simulated.

Viva Questions

1. What is the role of a post processor?
2. Difference between G00 and G01?
3. What is G54?
4. Why is tool simulation required before machining?

Experiment 5: Stress/Strain Evaluation of a Plate with a Hole

Aim

To evaluate the stress and strain distribution in a rectangular plate with a central circular hole under axial tension using FEA software (CAEFEM / ANSYS / CATIA Analysis), and compare with the analytical stress concentration.

Theory

A hole in a plate under tension disturbs the uniform stress flow, so stress rises near the hole edge. The **stress concentration factor** is $K_t = \sigma_{max} / \sigma_{nom}$. For a small hole in a wide plate $K_t = 3$. For a finite-width plate (width w, hole diameter d, thickness t, load P):

$\sigma_{nom} = P / ((w - d)\, t)$ $\quad K_t \approx 3.0 - 3.14(d/w) + 3.667(d/w)^2 - 1.527(d/w)^3$

Strain from Hooke's law: $\varepsilon = \sigma / E$. The maximum stress occurs at the hole edge on the section perpendicular to the load.

Procedure

1. Create the plate geometry (e.g. 100 mm x 50 mm x 5 mm with a 10 mm dia. central hole); using symmetry, a quarter model may be used.
2. Assign material: steel, E = 200 GPa, Poisson's ratio = 0.3.
3. Mesh with 2D plane-stress (or 3D solid) elements; refine the mesh near the hole.
4. Apply boundary conditions: fix one end; apply a tensile load (e.g. P = 10 kN, or equivalent pressure) on the other end.
5. Solve the model.
6. Plot von Mises stress, maximum principal stress and strain contours; note maximum values and location.
7. Compare the FEA σ_{max} with the analytical K_t x σ_{nom}.

Worked analytical check (for the above data)

Quantity	Value
w, d, t	50 mm, 10 mm, 5 mm
Load P	10 kN
$\sigma_{nom} = P/((w-d)t)$	10000 / (40 x 5) = 50 MPa
d/w	0.2
K_t	3.0 - 0.628 + 0.1467 - 0.0122 = 2.51
$\sigma_{max} = K_t\, \sigma_{nom}$	approx. 125.7 MPa
Strain at max = σ/E	125.7 / 200000 = 6.3×10^{-4}

The plate width in the worked check is 50 mm. Record your own FEA values in the table below.

Parameter	Analytical	FEA	% Error
Max stress (MPa)			
Max strain			
Location	Hole edge		

Result

The stress concentration at the hole was found and the FEA result agrees with the analytical value within acceptable error.

Viva Questions

1. What is stress concentration?
2. Why is the mesh refined near the hole?
3. Plane stress vs plane strain?
4. Where does failure usually start in a plate with a hole?